

SECTION 27 00 00

COMMUNICATIONS - GENERAL

PART 1 GENERAL

1.1 OVERVIEW

- A. Contractor is expected to furnish and install a complete and operational network cabling infrastructure (cable and connectivity) for Technology-related systems. Connectivity includes jacks, modular plugs, copper and fiber patch panels, fiber optic connectors, pre-terminated copper and fiber optic patch cords, and fiber optic insert panels (couplers for panels), and faceplates. Cable would include the fiber optic cable used for backbone runs to HS and Copper Category cable used for cameras, IP speakers, and Wireless Access Points.
 - 1. This also includes furnishing and installing networking equipment (network switches) and associated UPS (uninterruptible power supplies) to support the equipment plugged into the switches (IP Clock/Intercom Speakers, Wireless Access Points, computer workstations or VoIP phones, and surveillance cameras).
 - a. Due to exact loads not being known, contractor shall use the wattage of the power supply for a given switch when calculating run time.
 - 2. The Video Surveillance System cameras are required to be installed on a parallel network with separate fiber optic strands in the backbone connecting to a different switch in the ER at the HS. Coordinate with Owner for configuration information, IP address schemes and ranges, VLANs, and which switch and switch port(s) to use back at the HS. Contractor shall not patch cameras into same switch being used for the IP speakers, wireless access points, or workstation/VoIP ports.
- B. Contractor shall furnish and install all fiber optic patch cords and copper patch cords of the appropriate length, and match the performance of the patch panels and horizontal copper or backbone fiber optic cable and connector type being interconnected.

1.2 RELATED SECTIONS

- A. Division 26 - Electrical.

1.3 REFERENCES

- A. Institute of Electrical and Electronics Engineers (IEEE):
 - 1. 802.3 Ethernet Standards.
- B. TIA-569
- C. TIA-607
- D. TIA-758
- E. TIA-606
- F. TIA-568

1.4 SUBMITTALS

- A. Shop Drawings: Include details of construction, interface of equipment, and relationship with adjacent construction.
 - B. Product Submittal: Include Compliance letter along with detailed specifications of each product.
 - 1. Submit a front sheet index of part numbers and descriptions for easy review and cross reference during submittal review. Supporting specifications for each item should be presented in the same order within the submittal documents.
- 1.5 DELIVERY, STORAGE, AND HANDLING
- A. Deliver and store products in manufacturer's unopened packaging bearing the brand name and manufacturer's identification until ready for installation.
 - B. Handling: Handle materials to avoid damage.
- 1.6 PROJECT CONDITIONS
- A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's recommended limits.
- 1.7 SEQUENCING
- A. Ensure that products of this section are supplied to affected trades in time to prevent interruption of construction progress.
- 1.8 WARRANTY
- A. Provide all manufacturer warranties for performance after installation.

PART 2 PRODUCTS

2.1 J-HOOKS

- A. Contractor shall use metallic J-Hooks manufactured by Erico Caddy (nVent).
 - 1. J-hooks shall be installed no farther than 5'-0" apart from each other to properly support the cabling.
 - 2. Install to building structure, not ceiling grid stringers, lighting support wires, or other pathways.

2.2 CONDUIT

- A. Use conduit for cabling pathway over hard lid spaces or rooms where there is no ceiling and it is open to structure above.
- B. Conduit installation shall be compliant with TIA-569.
- C. Underground conduit installation and maximum distances shall be compliant with TIA-758.

2.3 INNERDUCT

- A. Innerduct shall be installed for the purpose of providing a continuous, protected cable route for fiber optic cable whenever fiber optic cable exits the conduit sleeve in a room. Minimum

innerduct sizing shall be one (1") inch. Innerduct shall not be considered a substitute for conduit or conduit sleeves when required by specification, drawings or code.

2.4 FIBER OPTIC CABLE

- A. Approved multimode fiber optic cables meet the Laser Optimized Multimode Fiber specifications in ANSI/TIA 492AAAF (OM4).

2.5 CATEGORY 6A CABLE

- 1. Cable shall be Category 6A ANSI/TIA-568.2-D compliant cable.

2.6 PLYWOOD BACKBOARD

- 1. Provide and vertically install wall-mounted, fireproofed, ACX, 4'x8' sheets of plywood backboard (Treated Plywood Terminal Board) on the walls of the IT Room as shown in the drawings. Contractor shall install backboard with the bottom edge of the backboard at +12" A.F.F. See electrical detail drawings for additional information regarding the total quantity of plywood backboard to be installed.

2.7 FACEPLATES

- A. Use single gang plastic faceplates with integrated label windows for Category 6A jack locations.

2.8 JACKS

- A. Use Category 6A jacks in faceplates as shown in the drawings.

PART 3 EXECUTION

3.1 INSTALLATION

- A. Category 6A Cabling.
 - 1. Cabling shall be installed above ceilings in conduit or j-hooks. Conduit pathways shall be required in rooms which have exposed structure above, or when cable has to pass over hard lid ceilings.
 - 2. Cabling in plenum spaces shall be rated appropriately.
- B. Backbone Underground
 - 1. Cabling shall be rated appropriately for OSP installation.
 - 2. Use factory conduit sweeps, with a minimum radius 6x the diameter of the conduit.
 - 3. Installation shall be compliant with TIA-758.
- C. Fiber Optic Cabling
 - 1. Terminate all multimode fiber optic cables as required with an LC connector. Land the LC connectors in the fiber interconnect center LC adapter plate.
 - 2. All fiber optic LC connectors shall be capable of achieving an insertion loss of .30 dB or less which is guaranteed under the warranty.
 - 3. Utilize outside plant rated cable whenever cable runs are routed through conduit under grade or outside the building envelope.
 - 4. Secure all cables at termination points to provide strain relief. Care shall be taken never to bend, crush, or kink cables. Support all fiber optic cables by their outer jacket. Never allow the fiber to be bent, hung, or supported in such a manner as to

provide the opportunity for breakage, microbends, or macrobends. Always maintain the highest quality wire management for all cable.

5. Cable routed underground shall be rated appropriately for OSP installation.
- D. Surveillance Cameras
 1. Install cameras and camera mounts provided by Owner.
 2. Install cable and cable pathway to each camera. Terminate end of cable at camera with an 8-pin, 8-position modular plug.
- E. Clock/Intercom IP PoE speakers
 1. Install cable and pathway to each speaker. Terminate end of cable at speaker with an 8-pin, 8-position modular plug.

3.2 COORDINATION

- A. Regardless of progress of overall installation, the network switches shall not be installed until the following is completed in the new IT room:
 1. Power – finished, tested, no further planned outages, and all outlets operational.
 2. Climate controlled – HVAC for room commissioned and functioning properly 24/7 with no further planned outages. Temperature in room shall be maintained at a constant level required by Owner.
 3. Cabling is installed and tested with passing results.
 4. All labeling has been completed.
 5. All cabinets, power supplies, PDUs, and UPS has all been installed.
 6. Permanent lighting for the room is installed and functioning correctly.
 7. Room can be secured and can be considered protected/secured. Doors to room must be installed.
 8. Room is sealed – wall penetrations, open ceiling tiles, underground conduit sleeve openings have been sealed to prevent ingress of vermin.
 9. All wall and floor finishing has been completed.
 10. Room has been thoroughly cleaned, with no equipment obstructing access to front and rear of racks. All wall mounted equipment and brackets shall be wiped clean of any dust or debris.
- B. Provide estimated dates equipment will be required so Owner can purchase cameras and wireless access points and schedule resources far enough in advance to avoid any conflicts or delays. Coordination should not begin until after shop drawings have been approved.
- C. For any existing equipment being relocated/reused, dates of outage as equipment is moved from old location to new location must be coordinated in conjunction with Architect and Owner.
 1. Be aware of when operation of Public Address loudspeakers are expected by the school.
 2. Attempt to make relocation occur over school holidays, provided it will be operational when staff and students return.
 - a. All testing and adjusting must also be complete.

3.3 LABELING

- A. All cable terminations shall be clearly and permanently labeled with appropriate cable numbering. Labeling of outlet jacks shall be provided on clear labels. Labeling of

termination system shall be provided on white labels. All labeling shall be black typed lettering. Handwritten labels shall not be permitted.

1. Contractor required to label at every location, using a printed label with adhesive intended to secure to the surface being used.
 2. Label all bonding and grounding conductors and busbars, indicating the bonding point on the other end of the cable.
 3. Label all faceplates and patch panels.
 4. Label all cables 6 inches (6") from the connector/termination end and at the patch panel or equipment termination. Do not use "flag" labels, but instead adhere along the length of the cable.
- B. Labeling shall comply with TIA-606.
- C. Cable Labeling shall identify the room number on the opposite end of the cable.
- D. Labels shall be machine generated/printed. No written labels shall be permitted. The Contractor shall be responsible for providing labels which have the appropriate adhesive, based on the surface being used.

3.4 TESTING

A. GENERAL

1. The Contractor shall resolve all tests that result in a FAIL prior to submitting to Owner for acceptance. Troubleshoot and resolve any failures. Most failures attributed to construction are going to occur at excessive bending points or at the connector.
2. Submit test reports with summary report in pdf format, including all labels from each test.
3. Submit full test report for each in pdf format.
4. Native file test reports shall be retained and provided to the A/E upon request if there are discrepancies or anomalies discovered in the test reports.
5. Testing shall conform to ANSI/TIA-568.0-E standard.
6. The structure for commercial building cabling is based on the generic cabling system structure specified in TIA-568-0.
7. Performance and technical criteria for balanced twisted-pair cabling systems are specified in TIA-568.2.
8. All testing shall be performed using the Fluke Versiv Family of Cabling Testing products. (DSX 8000/5000 for Copper and CFP/OFP Quad for fiber or above).
9. A factory calibration report shall show calibration performed within twelve months of the test report date and shall be included with the cable testing report package.
10. Provide UTP Category 6A and Optical Fiber OTDR Test Reports.
11. Provide Optical Fiber Power Loss Test Reports.
12. Test results which report a PASS*, FAIL*, or FAIL result for any of the parameters shall not be accepted. It is incumbent upon the Contractor to determine the cause of the issue, perform corrective action, and retest prior to submittal of test reports to the owner. The Contractor shall coordinate directly with the manufacturer if they are unable to resolve the issue in the field.

B. CATEGORY 6A CABLE

1. Upon completion of installation/termination of all cabling, the Contractor shall perform a 'Permanent Link' Certification test on each Cat6A cable to full ANSI/TIA 568-C compliance requirements.
2. Test results with incorrect NVP configured will be rejected and the Contractor will have to retest all cables. Verify correct NVP before testing. Remember that CMP,

CMR, and OSP cables each typically have a different NVP.

3. For cables where a modular plug is being used on the end for a device, such as camera or speaker, Contractor shall perform a Modular Plug Terminated Link (MTPL) test, compliant with TIA-568.2-D.
4. Each Test Report shall have the following information (verify it is correct prior to testing and submittal to Owner):
 - a. Cable ID (Label number)
 - b. Cable type being tested (use exact cable being used, select from manufacturer selection menu preconfigured in tester). If manufacturer is not listed within the database for the tester, use "generic" testing parameters with correct NVP and manually type in the cable type and part number. Verify with manufacturer and update tester for all current manufacturers before attempting a "generic" test. Also contact cable manufacturer to verify if part number may be identified under a previous company name/part number.
 - c. Testing of outlets shall only occur after the jacks are permanently installed/seated in their corresponding faceplate, junction box, or raceway. Contractor shall notify the Architect in writing of any locations which can't be permanently installed until the time of network integration or demolition of existing systems due to an obstruction. Testing prior to permanent installation on new construction projects is never permitted. Cables can be pinched, snagged, or compressed on the back of jacks during final placement of outlets into j-boxes, thereby causing pairs to be pulled free of termination points and create unwarranted delays on the project.
5. Provide three (3') feet service loops for each cable, coiled and properly supported per ANSI/TIA 568.0 requirements above the ceiling at each individual drop location.
6. Bend radius of all 4-pair copper wiring shall be no more than four (4) times the outside diameter of the cable under load and no-load conditions.
7. Cable pairs shall not be untwisted more than one quarter inch (¼") from the point of termination on all UTP four pair cables.
8. Cable sheath shall not be removed more than three quarters of an inch (¾") from the point of termination on all UTP four pair cables.

C. FIBER OPTIC OM4 MULTIMODE CABLE

1. Tests of multimode fiber optic cable shall be conducted utilizing an OTDR (Optical Time Domain Reflectometer with an integrated optical light source and optical power meter) at 850nm and 1300nm with the end-to-end attenuation in db/km recorded for each fiber.
2. Contractor shall perform Tier I and Tier II testing of fiber optic cable using the Fluke CertiFiber Pro OLTS (Optical Loss Test Set) for length and loss, and the OptiFiber Pro OTDR (Optical Time Domain Reflectometer) to show reflection of any imperfections or major "events" which contribute to loss. Maximum allowable fiber optic LC connector insertion loss (mated pair of connectors) shall be .30dB.
 - a. Test wavelengths in both directions, using one-cord reference method per ANSI/TIA-526-14-C-2015. This testing procedure is also referred to as the one-jumper reference method.
 - b. Contractor shall use reference grade patch cords from manufacturer of the cable being used or those recommended by the manufacturer of the tester.
 - c. Contractor shall only use the encircled flux test jumpers. Do not use mandrels to set reference for multimode testing.
 - d. The encircled flux launch controller from Fluke is preferred in order to guarantee the mandrel procedure is not inadvertently skipped by technicians in

the field and to ensure accuracy and consistency.

END OF SECTION